Algebra 2 Final Project - Semester 2

Standard	4 — Exceeds Expectations	3 — Meets Expectations	2 — Approaching Expectations	1 — Beginning
1. Sequences and Series	Selects and justifies appropriate sequence and series models using strong mathematical reasoning. Clearly distinguishes arithmetic vs geometric, recursive vs explicit, finite vs infinite, and discrete vs continuous representations when relevant. Explains what terms, parameters, and cumulative sums mean in context. Predictions, recalculations, and interpretations are thoughtful, realistic, and well-supported.	Uses appropriate sequence and series models with generally accurate reasoning. Explains arithmetic vs geometric patterns and recursive vs explicit forms with minor errors or missing detail. Interpretations and predictions are mostly reasonable and connected to context.	Shows partial understanding of sequences and series. Some model choices or explanations are inaccurate, incomplete, or weakly justified. Connections between formulas, graphs, and context may be unclear or inconsistent.	Shows limited understanding of sequences and series. Models are inappropriate, incomplete, or unsupported. Explanations and interpretations are mostly missing or incorrect.
2. Exponentials and Logarithms	Selects and justifies exponential models appropriately. Clearly explains growth and decay, thresholds, growth factors, and logarithmic reasoning in context. Thoughtfully compares exponential behavior to linear or geometric behavior when appropriate. Solutions are interpreted realistically with strong contextual understanding and critique of limitations.	Uses exponential and logarithmic reasoning appropriately with mostly accurate explanations. Shows general understanding of growth factors, thresholds, and logarithmic solving. Interpretations are usually reasonable with only minor errors or missing detail.	Shows partial understanding of exponentials and logarithms. Makes multiple errors in solving, interpretation, or model choice. Explanations may be vague, inconsistent, or disconnected from context.	Shows limited understanding of exponential or logarithmic reasoning. Work is incomplete, incorrect, or lacks meaningful interpretation.
3. Polynomials	Creates and analyzes polynomial models with strong mathematical reasoning. Clearly explains degree, leading coefficient, end behavior, intercepts, multiplicity, extrema, and realistic domain restrictions in context. Thoughtfully compares factored and standard form and explains what each representation reveals. Uses notation such as $x \to \infty$ and $x \to -\infty$ correctly and meaningfully.	Analyzes polynomial models accurately with generally clear explanations. Shows understanding of graph behavior, structure, and polynomial forms with only minor errors or missing detail. Most interpretations are connected to context.	Shows partial understanding of polynomial structure and graph behavior. Multiple errors or incomplete explanations weaken interpretation and analysis. Contextual meaning may be unclear or inconsistent.	Shows limited understanding of polynomial reasoning or graph behavior. Analysis is incomplete, incorrect, or unsupported by meaningful explanation.

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4. Mathematical Communication and Interpretation (Skill Grade)	Mathematical thinking is communicated clearly and thoughtfully through explanations, graphs, annotations, tables, discussion, and contextual interpretation. Strong connections among graph, algebra, assumptions, and context are consistently visible. Reasoning is organized, precise, and easy to follow.	Communication is generally clear and understandable. Most mathematical ideas are connected to graphs, algebra, or context appropriately, though some explanations may lack precision or depth.	Explanations are inconsistent, incomplete, or difficult to follow. Connections between mathematics and context are weak, unclear, or only partially developed.	Mathematical thinking is minimally communicated. Explanations, interpretations, and connections are mostly missing or unclear.
5. Comparing Models, Assumptions, and Limitations (Skill Grade)	Thoughtfully compares multiple models, assumptions, and representations such as discrete vs continuous, recursive vs explicit, linear vs exponential, or different polynomial structures. Critiques realism, hidden variables, tradeoffs, assumptions, and unreasonable predictions with strong insight. Revision strengthens the mathematical investigation in meaningful ways.	Compares models and assumptions appropriately with some explanation of limitations or realism. Shows general understanding of how assumptions affect predictions and interpretations. Some revision or reconsideration of ideas is visible.	Comparisons are limited, superficial, or weakly justified. Explanations of assumptions, realism, or limitations are incomplete or inconsistent. Revision is minimal or mostly procedural.	Shows little or no meaningful comparison of models, assumptions, or limitations. Revision and critique are mostly absent.
6. Algebraic Precision and Flexible Number Sense (Skill Grade)	Solves equations accurately using appropriate algebraic methods such as factoring, logarithms, inverse operations, substitution, graph interpretation, and algebraic manipulation. Fractions, decimals, percentages, negative numbers, and units are used fluently and precisely. Work is organized, justified, and consistently checked for reasonableness.	Uses appropriate algebraic methods with mostly accurate calculations and organization. Rational numbers and units are generally used correctly, with only minor errors or omissions. Most answers are reasonable and interpreted appropriately.	Uses some correct methods but makes multiple algebraic, numerical, or organizational errors. Precision, units, or interpretation may be inconsistent or incomplete.	Shows little or no valid algebraic progress. Calculations, precision, organization, or interpretation are mostly incorrect or missing.